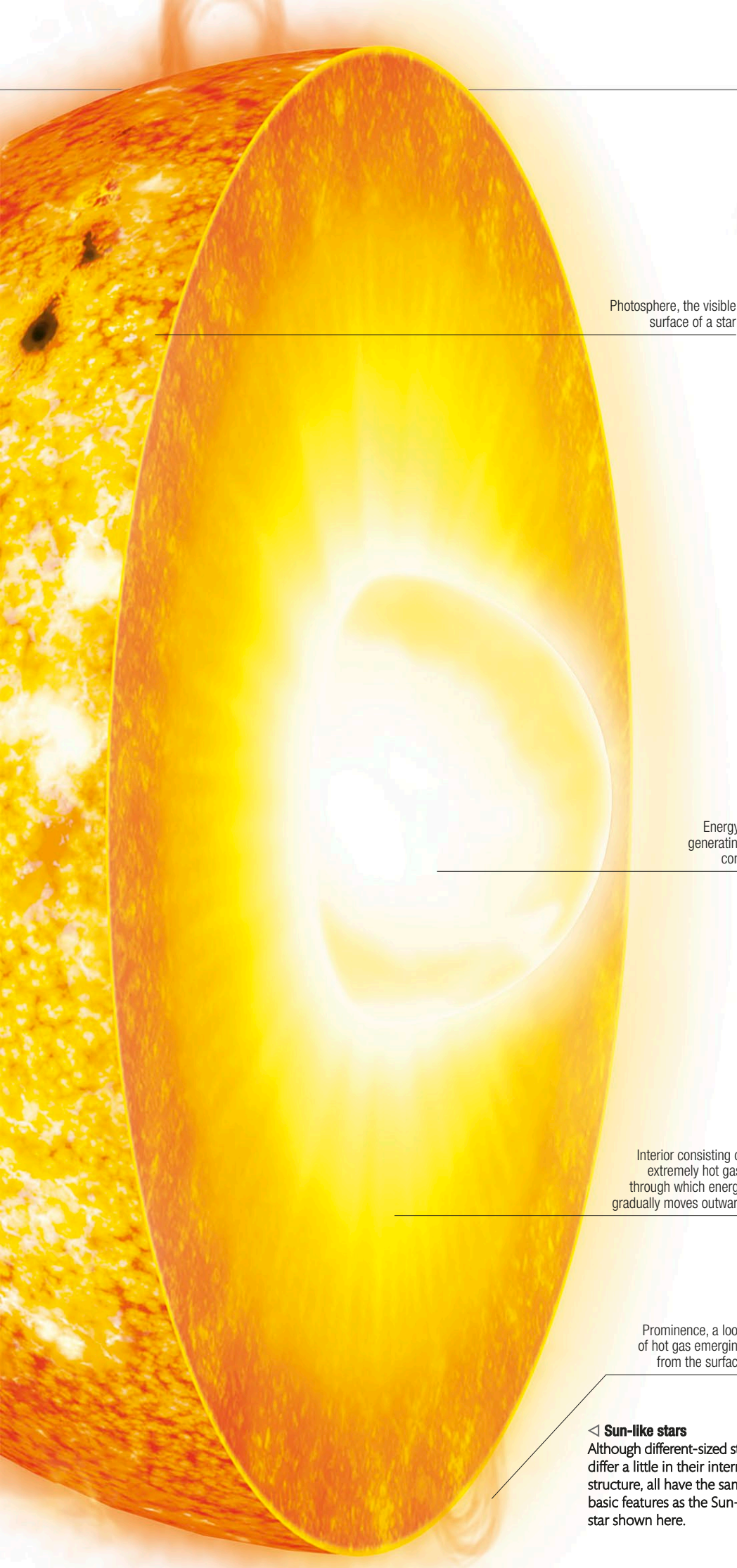




WHAT IS A STAR?

A STAR IS AN ENORMOUS BALL OF EXTREMELY HOT GAS THAT PRODUCES ENERGY IN ITS CORE AND EMITS THIS ENERGY AT ITS SURFACE.

All the individual stars we can see in the night sky are part of our own galaxy, the Milky Way. Although in cosmic terms these are all “local” stars, they are actually fantastically far away—the closest is nearly 25 trillion miles (40 trillion km) distant, and most are much farther off. Overall in our galaxy there are more than 200 billion stars, of which about 10,000 are visible to the naked eye.

Star appearance and variation

We see all stars in the night sky as just tiny pinpricks of light. Some look brighter than others, but with the unaided eye they don't seem to differ much in color: all look rather white. In fact, stars are much more varied than might at first appear. They come in a vast range of sizes and temperatures, in an array of colors, and also differ greatly in age and life span. Many of these characteristics of stars are related. For example, a star's surface temperature and color are closely linked—a star with a relatively low surface temperature glows red, whereas hotter stars appear (with increasing temperature) orange, yellow, white, or blue.

SPECTRAL CLASSIFICATION OF STARS

Class	Apparent colour	Average surface temperature	Example star
 O	Blue	over 54,000°F (30,000°C)	Zeta Puppis, also called Naos (Puppis)
 B	Deep blueish white	36,000°F (20,000°C)	Rigel (Orion)
 A	Pale blueish white	15,000°F (8,500°C)	Sirius A (Canis Major)
 F	White	11,700°F (6,500°C)	Procyon A (Canis Minor)
 G	Yellow-white	9,500°F (5,300°C)	The Sun
 K	Orange	7,150°F (4,000°C)	Aldebaran (Taurus)
 M	Red	5,350°F (3,000°C)	Betelgeuse (Orion)

◀ Sun-like stars

Although different-sized stars differ a little in their internal structure, all have the same basic features as the Sun-like star shown here.

△ Star spectral classes

The spectrum of light from a star carries a lot of information about the star. By studying its spectrum, scientists can assign any star to a type, called a spectral class, of which the main ones are listed above.